

Forward Reasoning in Isabelle/HOL

```
theory FW
imports Main
begin

lemma implication_introduction:
  " $X \implies (Y \longrightarrow X)$ "
  sorry

thm implication_introduction

lemma neg_intro:
  " $(X \longrightarrow Y) \implies (X \longrightarrow \neg Y) \implies \neg X$ "
  sorry

thm neg_intro

lemma assumes
  assum1: " $P \longrightarrow Q$ " and assum2: " $Q \longrightarrow R$ " and assum3: " $\neg R$ "
  shows " $\neg P$ "
proof-
  have 1: " $\neg R$ "
    using assum3
  .
  have 2: " $Q \longrightarrow R$ "
    using assum2
  .
  have 3: " $Q \longrightarrow \neg R$ "
    using implication_introduction[OF 1]
  .
  have 4: " $\neg Q$ "
    using neg_intro[OF 2 3]
  .
  have 5: " $P \longrightarrow \neg Q$ "
    using implication_introduction[OF 4]
  .
  show ?thesis
    using neg_intro[OF assum1 5]
  .
qed

end
```